



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

#### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

|                             |                                                |                     |
|-----------------------------|------------------------------------------------|---------------------|
| <b>TITLE OF THE PROJECT</b> | <b>AI-Adaptive Blood Donor Matching System</b> |                     |
| <b>S. No.</b>               | <b>University ID</b>                           | <b>Name</b>         |
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| <b>Name of the Guide</b>    |                                                | Dr. Archana Kalindi |

### Abstract

The **AI-Adaptive Blood Donor Matching System** is a real-world application designed to help hospitals and blood banks quickly identify suitable blood donors during emergencies. The system analyzes blood requirements and donor information such as blood group compatibility, location, availability, donation history, and emergency priority. An adaptive matching mechanism generates compatibility and priority scores to recommend suitable available donors and respond to changes in donor availability and blood requirements.

The system uses **Machine Learning and Natural Language Processing (NLP)** to improve the donor matching and recommendation process. **React.js** is used to develop the frontend, while **FastAPI and Python** are used for backend development. **MongoDB** is used for data management, and **REST APIs and WebSockets** are used for communication and real-time updates. **Docker** is used for containerization, and **GitHub** is used for version control and progressive project development.

The system aims to provide an efficient and user-friendly platform for hospitals, blood banks, and registered donors. By intelligently matching and prioritizing suitable donors based on multiple factors, the project aims to reduce the time required to identify blood donors during emergencies and improve the overall efficiency of blood donation management.

Signature of the Guide

HOD